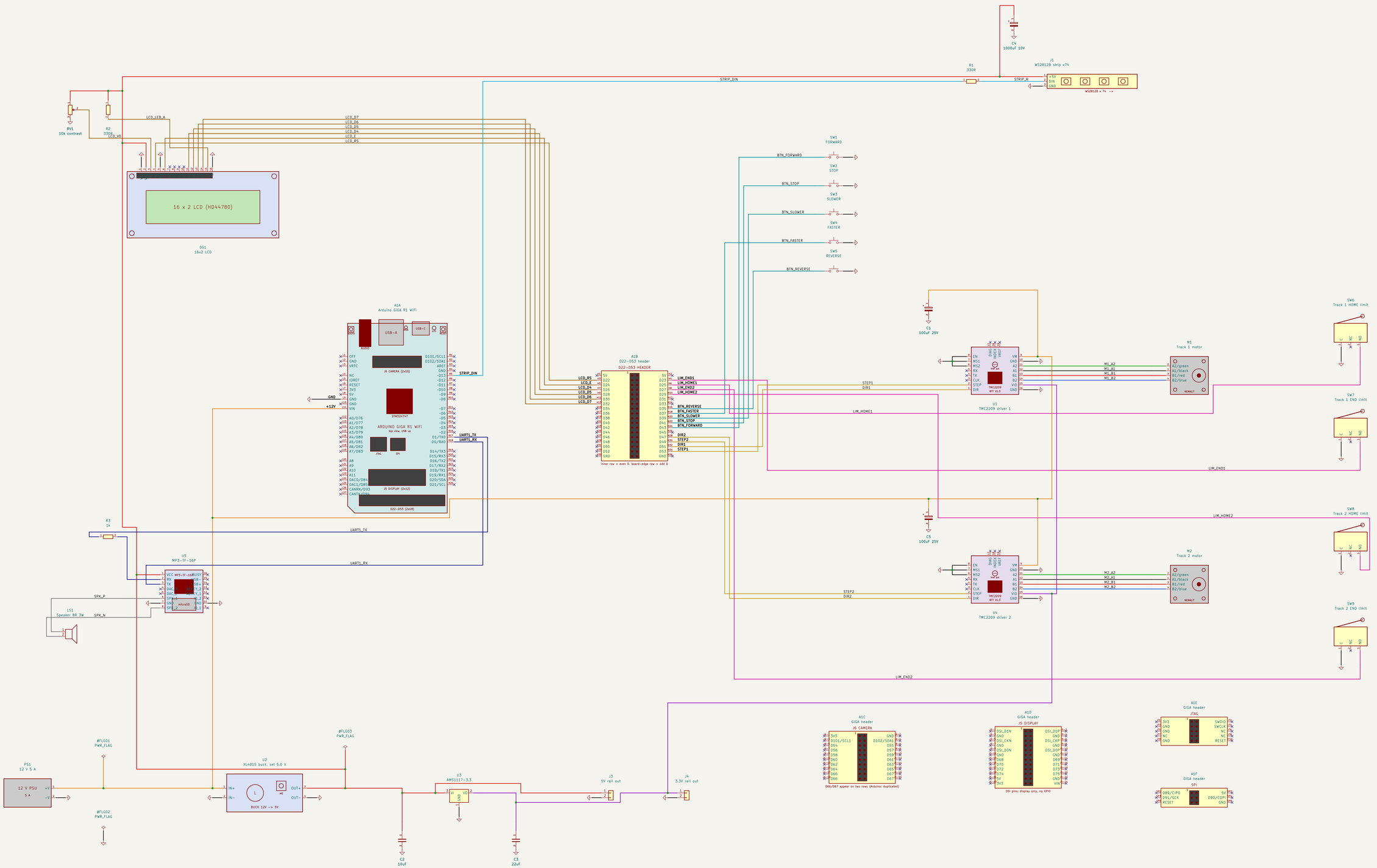
CARNIVAL GAME — FULL WIRING DIAGRAM (every part, every pin, every connection)

Every line is a real wire between two pins; A dot is a joint; lines that cross without a dot do not touch.
GROUND is drawn as the ground symbol at each pin (so ground wire runs round the sheet); every ground symbol is the same connection.
Parts are drawn to look like the real hardware: the GIGA board (top view, USB up) with every header pin, the TMC2209 StepSticks, NEMA17 motors,
lever microswitches, the 16x2 LCD module, the MP3-TF-16P module, the WS2812B strip, the buck module and the 12 V PSU. X = pin not used.
WIRE COLOURS motor coil A: A1 black, A2 green / coil B: B1 red, B2 blue | +12V orange +5V red +3V3 purple GND black | buttons teal LCD brown limit switches pink
STEP/DIR gold MP3 serial navy LED data cyan speaker grey. Every used GIGA pin has its signal name printed on its lead.
Not yet confirmed on hardware: Track 1 limits 25/23 (guessed); Track 2 LCD and MP3 pins proposed. Confirmed 2026-09-30: buttons 35-43, Track 1 STEP 53 / DIR 51.



Matches tests/test_track_homing/track_homing.ino
Carnival Game — FULL wiring diagram (every pin, every wire)
SBU VIP Bear Carnival
Sheet: /
File: carnival_full.kicad_sch

Title: Full wiring diagram

Size: A1	Date: 2026-09-30	**Rev: E**
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